

## 14.9 The Determinant as a Top Exterior Power

Chapter 6 built the determinant twice. First it proved that an alternating  $n$ -linear form on  $F^n$  is determined by its value at the standard basis (Theorem 6.4.2), which says *at most one* normalized such form exists; then it wrote the Leibniz sum down and checked the three rules (Theorem 6.4.3), which says *at least one* does. The number  $\det \mathbf{A}$  of Definition 6.4.1 is that form's value. This section builds the determinant a third time, and the honest way to describe what it adds is to say up front what it does not: no new value, no faster computation, and no theorem Chapter 6 was missing. What it adds is an **object**. The exterior powers of the previous section turn  $\det T$  from a number attached to a matrix into the operator that  $T$  induces on a one-dimensional space, and from that description multiplicativity stops being a computation and becomes the statement that composing two induced maps induces the composite.

Throughout,  $F$  is a field,  $V$  and  $W$  are finite-dimensional vector spaces over  $F$ , and  $\Lambda^k V$ , the wedge  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$ , and the universal property for alternating maps are as in Definition 14.8.2. We use freely that the wedges  $\mathbf{v}_{i_1} \wedge \cdots \wedge \mathbf{v}_{i_k}$  over strictly increasing index tuples form a basis of  $\Lambda^k V$ , so that  $\dim \Lambda^k V = \binom{n}{k}$  when  $\dim V = n$  (Theorem 14.8.5), and that a wedge of vectors is non-zero exactly when the list is linearly independent (Theorem 14.8.7).

### 14.9.1 Wedging with a fixed vector

Two facts from the previous section will be used on every page, and one small tool is still missing.

The first fact is the **reordering rule**: the wedge map  $(\mathbf{v}_1, \dots, \mathbf{v}_k) \mapsto \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$  is alternating (Definition 14.8.2), and an alternating multilinear map satisfies

$$f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = \operatorname{sgn}(\sigma) f(\mathbf{v}_1, \dots, \mathbf{v}_k)$$

for every  $\sigma \in S_k$ , by Lemma 14.8.1 (b). Reordering a wedge costs the sign of the permutation, and nothing else. The second is that  $\Lambda^n V$  is one-dimensional when  $\dim V = n \geq 1$  (Corollary 14.8.6); that is the whole engine of this section.

The missing tool is the ability to wedge a fixed vector onto a whole exterior power, not just onto a list of vectors. Section 10 builds the product on all of  $\Lambda V$ ; here one vector at a time is enough.

#### Lemma 14.9.1 — Wedging with a Fixed Vector

Let  $\mathbf{v} \in V$  and  $k \geq 1$ . There is a **unique** linear map

$$\varepsilon_{\mathbf{v}}: \Lambda^k V \rightarrow \Lambda^{k+1} V, \quad \varepsilon_{\mathbf{v}}(\mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_k) = \mathbf{v} \wedge \mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_k.$$

*Proof.* The map  $V^k \rightarrow \Lambda^{k+1} V$  sending  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  to  $\mathbf{v} \wedge \mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_k$  is  $k$ -linear, because the wedge of  $k+1$  vectors is linear in each slot and the first slot is frozen at  $\mathbf{v}$ ; and it is alternating, because if  $\mathbf{w}_p = \mathbf{w}_r$  with  $p \neq r$  then the wedge of  $k+1$  vectors has two equal arguments and is  $\mathbf{0}$ . By the universal property in Definition 14.8.2 it factors through  $\Lambda^k V$ , giving a linear  $\varepsilon_{\mathbf{v}}$  with the stated values. Uniqueness holds because the wedges  $\mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_k$  span  $\Lambda^k V$  (Theorem 14.8.5), and a linear map is determined on a spanning set.

## 14.9.2 Induced maps on exterior powers

A linear map should act on everything built from its domain. A map  $T: V \rightarrow W$  sends a list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  to the list  $(T\mathbf{v}_1, \dots, T\mathbf{v}_k)$ , so it ought to send the wedge  $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$  to  $T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$ . The only question is whether that rule defines a map at all: an element of  $\Lambda^k V$  can be written as a wedge in many ways, and an assignment made on wedges must be checked. The universal property answers it in two lines, and that is the point of having a universal property.

*The  $k$ -th exterior power of a map wedges its values together.*

### Definition 14.9.1 — Exterior Power of a Linear Map

Let  $T: V \rightarrow W$  be linear and let  $k \geq 1$ . The  **$k$ -th exterior power** of  $T$  is the unique linear map

$$\Lambda^k T: \Lambda^k V \rightarrow \Lambda^k W$$

satisfying  $\Lambda^k T(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k) = T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$  for all  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ .

**Well-definedness.** The map  $V^k \rightarrow \Lambda^k W$  sending  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  to  $T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$  is  $k$ -linear, since  $T$  is linear and the wedge in  $W$  is linear in each slot; and it is alternating, since  $\mathbf{v}_p = \mathbf{v}_r$  forces  $T\mathbf{v}_p = T\mathbf{v}_r$  and a wedge with two equal arguments is  $\mathbf{0}$ . So Definition 14.8.2 produces exactly one linear map on  $\Lambda^k V$  with the stated values, and the wedges span  $\Lambda^k V$  (Theorem 14.8.5), so no other linear map has them.

**Examples.**

- $k = 1$ .  $\Lambda^1 V = V$  and  $\Lambda^1 T = T$ : the definition says nothing new in degree one.
- **The identity.**  $\Lambda^k(\text{id}_V) = \text{id}$ , since both sides send  $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$  to itself.
- **A scalar.** For  $T = c \text{id}_V$ ,  $\Lambda^k T(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k) = c^k \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$ , so  $\Lambda^k T = c^k \text{id}$ . One factor of  $c$  comes out of each of the  $k$  slots.
- **A degenerate case.** If  $k > \dim V$  then  $\Lambda^k V = \{\mathbf{0}\}$  (Theorem 14.8.5) and  $\Lambda^k T$  is the zero map between zero spaces. Worth recording so that statements below need no size caveat.

**Non-example by minimal change.** Replace “alternating” by nothing: the rule  $\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_k \mapsto T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$  on  $V^{\otimes k}$  is perfectly well defined, but it is **not** the map above, because  $\Lambda^k V$  is a quotient of  $V^{\otimes k}$  and not a subspace of it. What makes Definition 14.9.1 legal is precisely that the rule kills the relations that were divided out.

The reason for having the notation is the next sentence.

### Theorem 14.9.2 — Exterior Powers Respect Composition

Let  $T: U \rightarrow V$  and  $S: V \rightarrow W$  be linear, and let  $k \geq 1$ . Then

$$\Lambda^k(ST) = (\Lambda^k S)(\Lambda^k T), \quad \Lambda^k(\text{id}_V) = \text{id}_{\Lambda^k V}.$$

*Proof.* Both  $\Lambda^k(ST)$  and  $(\Lambda^k S)(\Lambda^k T)$  are linear maps  $\Lambda^k U \rightarrow \Lambda^k W$ , so it suffices to compare them on the wedges, which span  $\Lambda^k U$  (Theorem 14.8.5). By Definition 14.9.1 applied three times,

$$(\Lambda^k S)(\Lambda^k T)(\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k) = ST\mathbf{u}_1 \wedge \cdots \wedge ST\mathbf{u}_k = \Lambda^k(ST)(\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k).$$

The second identity was checked above. This proves the theorem. ■

**Example. (The second exterior power of a  $3 \times 3$  matrix)**

Let  $T: \mathbb{Q}^3 \rightarrow \mathbb{Q}^3$  have matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}$$

in the standard basis. Find the matrix of  $\Lambda^2 T$  in the basis  $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$  of  $\Lambda^2 \mathbb{Q}^3$ , and compute its determinant.

*Solution.* By Definition 14.9.1,  $\Lambda^2 T(\mathbf{e}_i \wedge \mathbf{e}_j) = \mathbf{a}_i \wedge \mathbf{a}_j$ , where  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$  are the columns of  $\mathbf{A}$ . Expand one column:

$$\begin{aligned} \mathbf{a}_1 \wedge \mathbf{a}_2 &= (2\mathbf{e}_1 + 5\mathbf{e}_3) \wedge (\mathbf{e}_1 + 4\mathbf{e}_2 + 2\mathbf{e}_3) \\ &= 8\mathbf{e}_1 \wedge \mathbf{e}_2 + 4\mathbf{e}_1 \wedge \mathbf{e}_3 + 5\mathbf{e}_3 \wedge \mathbf{e}_1 + 20\mathbf{e}_3 \wedge \mathbf{e}_2 \\ &= 8\mathbf{e}_1 \wedge \mathbf{e}_2 - \mathbf{e}_1 \wedge \mathbf{e}_3 - 20\mathbf{e}_2 \wedge \mathbf{e}_3, \end{aligned}$$

where the terms  $2\mathbf{e}_1 \wedge \mathbf{e}_1$  and  $10\mathbf{e}_3 \wedge \mathbf{e}_3$  were dropped in the second line because a wedge with a repeated vector is  $\mathbf{0}$ , and the third line used  $\mathbf{e}_3 \wedge \mathbf{e}_1 = -\mathbf{e}_1 \wedge \mathbf{e}_3$  and  $\mathbf{e}_3 \wedge \mathbf{e}_2 = -\mathbf{e}_2 \wedge \mathbf{e}_3$  (Lemma 14.8.1 (b)). The coefficients  $8, -1, -20$  are the three  $2 \times 2$  minors of the first two columns of  $\mathbf{A}$ , on rows  $\{1, 2\}, \{1, 3\}, \{2, 3\}$  — a pattern the last part of this section explains. Doing the same for the other two column pairs gives

$$[\Lambda^2 T] = \begin{pmatrix} 8 & 2 & -11 \\ -1 & -13 & -5 \\ -20 & -5 & 2 \end{pmatrix},$$

whose determinant is 2601. Since  $\det \mathbf{A} = -51$ , this is  $(\det \mathbf{A})^2$ .

*Quick check.* In the example, why is  $\Lambda^2 T$  **not** multiplication by a scalar, while  $\Lambda^3 T$  is?

*Solution.*  $\dim \Lambda^2 \mathbb{Q}^3 = \binom{3}{2} = 3$  by Theorem 14.8.5, and the displayed matrix is not a scalar multiple of  $\mathbf{I}_3$ , so  $\Lambda^2 T$  is an operator on a three-dimensional space with no reason to be scalar. But  $\dim \Lambda^3 \mathbb{Q}^3 = \binom{3}{3} = 1$ , and every linear operator on a one-dimensional space is multiplication by a scalar. Being a line is what makes the top power special.

### 14.9.3 The top exterior power

That last observation is the whole theorem. When  $k$  equals  $\dim V$ , the space  $\Lambda^k V$  is a line, an operator on a line is a scalar, and the scalar can be read off by feeding in a basis.

#### Theorem 14.9.3 — The Determinant Is the Top Exterior Power

Let  $V$  be a vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . Then  $\dim \Lambda^n V = 1$ , and  $\Lambda^n T$  is multiplication by  $\det T$ :

$$\Lambda^n T(\omega) = (\det T) \omega \quad \text{for every } \omega \in \Lambda^n V.$$

Equivalently, for all  $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$ ,

$$T\mathbf{v}_1 \wedge \cdots \wedge T\mathbf{v}_n = (\det T) \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n. \quad (14.9.1)$$

*Idea.* Three moves. ①  $\Lambda^n V$  is one-dimensional, so  $\Lambda^n T$  is multiplication by one unknown scalar  $c$ . ② To find  $c$ , evaluate on a basis wedge  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n$  and expand each  $T\mathbf{v}_j$  in the basis. ③ The expansion produces one term for every function from slots to basis indices; the non-injective ones die because a wedge with a repeat is zero, and what survives is a sum over permutations with signs — the Leibniz formula, arriving on its own.

*Proof.* Fix a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$ . By Corollary 14.8.6,  $\dim \Lambda^n V = 1$  with basis the single wedge  $\omega_0 = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n$ . Since  $\Lambda^n T$  is a linear operator on a one-dimensional space, there is a unique  $c \in F$  with  $\Lambda^n T(\omega_0) = c\omega_0$ , and then  $\Lambda^n T(a\omega_0) = ac\omega_0$  for every  $a \in F$ , so  $\Lambda^n T = c \text{ id}$  on all of  $\Lambda^n V$ .

It remains to show  $c = \det T$ . Let  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}}$  have entries  $a_{ij}$ , so that  $T\mathbf{v}_j = \sum_{i=1}^n a_{ij}\mathbf{v}_i$  (Definition 3.6.1). The wedge of  $n$  vectors is linear in each slot, so expanding slot by slot gives one term for each function  $\varphi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ :

$$T\mathbf{v}_1 \wedge \cdots \wedge T\mathbf{v}_n = \sum_{\varphi} a_{\varphi(1)1} \cdots a_{\varphi(n)n} \mathbf{v}_{\varphi(1)} \wedge \cdots \wedge \mathbf{v}_{\varphi(n)}.$$

If  $\varphi$  is not injective, two slots hold the same vector and the wedge is  $\mathbf{0}$ . A non-injective  $\varphi$  is possible only because the domain and codomain are finite of the same size, so the surviving  $\varphi$  are exactly the bijections, that is, the permutations  $\sigma \in S_n$ . For those, Lemma 14.8.1 (b) gives  $\mathbf{v}_{\sigma(1)} \wedge \cdots \wedge \mathbf{v}_{\sigma(n)} = \text{sgn}(\sigma) \omega_0$ . Therefore

$$\Lambda^n T(\omega_0) = \left( \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n} \right) \omega_0,$$

and the bracket is  $\det \mathbf{A}$  by Definition 6.4.1, which is  $\det T$  by Definition 6.5.1. So  $c = \det T$ .

For the last display, note that both sides of Eq. 14.9.1 are  $n$ -linear in  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  and agree when the list is the basis  $\mathcal{B}$ ; but in fact no extra argument is needed, since  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n \in \Lambda^n V$  for **any** list, and the first display applies to it. This proves the theorem. ■

**Warning. Only the top power is a line.** For  $1 \leq k < n$  the operator  $\Lambda^k T$  is not multiplication by a scalar in general: [Example \(The second exterior power of a  \$3 \times 3\$  matrix\)](#) exhibits a  $\Lambda^2 T$  with three different diagonal entries. The determinant appears because  $\binom{n}{n} = 1$ , not because exterior powers are scalars.

**Warning. A map between different spaces still has no determinant.** If  $T: V \rightarrow W$  is linear with  $\dim V = \dim W = n$ , then  $\Lambda^n T$  is a linear map between two **different** one-dimensional spaces, and there is no scalar attached to it until bases of  $\Lambda^n V$  and  $\Lambda^n W$  are chosen. Changing the basis of  $\Lambda^n W$  alone rescales the number arbitrarily. This is the same obstruction Chapter 6 recorded after Definition 6.5.1, now visible as the failure of a line to have a canonical generator.

**Quick check.** Let  $T$  be the operator on  $F^3$  with  $T\mathbf{e}_1 = \mathbf{e}_2$ ,  $T\mathbf{e}_2 = \mathbf{e}_3$  and  $T\mathbf{e}_3 = \mathbf{e}_1$ . Read  $\det T$  off [Eq. 14.9.1](#), without writing down a matrix.

**Solution.**  $T\mathbf{e}_1 \wedge T\mathbf{e}_2 \wedge T\mathbf{e}_3 = \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1$ . The permutation carrying  $(1, 2, 3)$  to  $(2, 3, 1)$  is a 3-cycle, of sign  $+1$  (Proposition 6.3.4), so by Lemma 14.8.1 (b) this equals  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$ . Comparing with [Eq. 14.9.1](#) on the non-zero element  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$  gives  $\det T = 1$ .

## 14.9.4 What the third construction adds

It is worth saying plainly, because the temptation to oversell a slick construction is real.

**It adds no new value.** The scalar produced in the proof is the Leibniz sum, arrived at by the same expansion that proved Theorem 6.4.2 in Chapter 6: expand multilinearly, discard the terms with a repeated basis vector, sort the survivors and collect signs. Anyone who has read Chapter 6 §04 has read that computation. The exterior power did not compute the determinant; it stored the same computation in a different place.

**It adds no faster method.** Nothing here helps evaluate a determinant. Elimination remains the way to do that (Example (A  $4 \times 4$  Determinant by Elimination)).

**What it does add is that  $\det$  is a map, not just a number.** Chapter 6 defined  $\det \mathbf{A}$  by a formula in the entries and then had to *prove* that operators inherit it: the formula depends on a basis, so Corollary 6.5.7 was needed before Definition 6.5.1 could be written down. Here  $\Lambda^n T$  is defined with no basis in sight, and the scalar exists because  $\Lambda^n V$  is a line. Basis-independence is not a theorem about the determinant; it is the absence of any choice to be independent of. One honest qualification: the route is not self-contained. Theorem 14.8.5, which is what makes  $\Lambda^n V$  a line in the first place, proves independence with a  $k \times k$  determinant, and the identification of the scalar with  $\det T$  goes through Definition 6.5.1. Chapter 6 is not being replaced here; it is being re-read.

And from that, multiplicativity is free.

### Corollary 14.9.4 — Multiplicativity and Invertibility, from Functoriality

Let  $\dim V = n \geq 1$  and  $S, T \in \mathcal{L}(V)$ .

- (a)  $\det(ST) = \det S \det T$ , and  $\det \text{id}_V = 1$ .
- (b)  $T$  is invertible if and only if  $\det T \neq 0$ .

*Proof.*

- (a) By Theorem 14.9.2,  $\Lambda^n(ST) = (\Lambda^n S)(\Lambda^n T)$ . By Theorem 14.9.3 all three maps are multiplications by scalars on the line  $\Lambda^n V$ , namely by  $\det(ST)$ ,  $\det S$  and  $\det T$ ; and multiplication by  $\det T$  followed by multiplication by  $\det S$  is multiplication by  $\det S \det T$ . Two scalar multiplications on a non-zero space are equal only if the scalars are, so  $\det(ST) = \det S \det T$ . Likewise  $\Lambda^n(\text{id}_V) = \text{id}$ , so  $\det \text{id}_V = 1$ .
- (b)  $(\Rightarrow)$  If  $T$  is invertible, apply (a) to  $TT^{-1} = \text{id}_V$ : then  $\det T \cdot \det(T^{-1}) = 1$ , so  $\det T \neq 0$ .  $(\Leftarrow)$  Suppose  $T$  is not invertible. Then  $T$  is not injective, so  $\ker T \neq \{0\}$ ; choose  $\mathbf{v}_1 \in \ker T$  with  $\mathbf{v}_1 \neq 0$  and extend to a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  (Theorem 1.6.5). Since  $T\mathbf{v}_1 = 0$  and the wedge is linear in its first slot,  $T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_n = 0$ . By Eq. 14.9.1 this is  $(\det T) \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_n$ , and the wedge of a basis is non-zero (Theorem 14.8.7), so  $\det T = 0$ .

■

Part (b) is the promise Theorem 14.8.7 made: the test “is this list independent?” and the test “is this determinant non-zero?” are the same test, once the determinant is read as an induced map.

Compare the work this replaces. Chapter 6 proved Theorem 6.5.4 by fixing  $\mathbf{A}$ , checking that  $\mathbf{B} \mapsto \det(\mathbf{AB})$  is alternating and  $n$ -linear in the columns of  $\mathbf{B}$ , and applying uniqueness. That proof is short and correct; what the present one shows is that the content is not really about determinants at all. Multiplicativity is the statement that  $\Lambda^n$  turns composition into composition — and  $\Lambda^n$  does that for the same reason that  $\Lambda^k$  does it for every  $k$ , where no scalar is available and no multiplicativity statement can even be phrased.

*Remark.* What is **not** improved: Theorem 6.4.5. The identity  $\det \mathbf{A}^\top = \det \mathbf{A}$  is a statement about rows versus columns, and  $\Lambda^n$  sees only the map  $T$ , which has no rows. Chapter 6’s proof, reindexing the Leibniz sum by  $\sigma^{-1}$ , remains the natural one.

## 14.9.5 The coordinates of a wedge

The example above produced the minors of a matrix as the coefficients of a wedge, apparently by accident. It was not an accident, and the general statement is the bridge between this chapter and Chapter 6 §07.

Fix  $n \geq 1$ , write  $[n] = \{1, \dots, n\}$  as Chapter 6 §07 does, and recall the notation of Definition 6.7.1: for  $\mathbf{A} \in M_{n \times k}(F)$  and a set  $I = \{i_1 < \dots < i_k\} \subseteq [n]$  of size  $k$ , the submatrix  $\mathbf{A}_{I,[k]} \in M_k(F)$  keeps the rows in  $I$  and all  $k$  columns, and its determinant is a  $k \times k$  minor of  $\mathbf{A}$ . Write

$$\mathbf{e}_I := \mathbf{e}_{i_1} \wedge \mathbf{e}_{i_2} \wedge \dots \wedge \mathbf{e}_{i_k} \in \Lambda^k F^n,$$

so that the  $\mathbf{e}_I$ , over all  $k$ -element  $I \subseteq [n]$ , form the basis of Theorem 14.8.5.

### Theorem 14.9.5 — The Coordinates of a Wedge Are the Minors

Let  $1 \leq k \leq n$ , let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^n$ , and let  $\mathbf{A} \in M_{n \times k}(F)$  be the matrix with these vectors as its columns. Then

$$\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \cdots \wedge \mathbf{v}_k = \sum_I (\det \mathbf{A}_{I,[k]}) \mathbf{e}_I,$$

the sum running over all subsets  $I \subseteq [n]$  with  $|I| = k$ . That is, the coordinate of  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$  at  $\mathbf{e}_I$  is the  $k \times k$  minor of  $\mathbf{A}$  on the rows  $I$ .

*Idea.* The same expansion as in Theorem 14.9.3, but now the index functions run into a bigger set. An injective  $\varphi: [k] \rightarrow [n]$  is a choice of image set  $I$  followed by an ordering of it, so the terms group into one bunch per  $I$ , and each bunch is a signed sum over  $S_k$ : a determinant of size  $k$ .

*Proof.* Write  $\mathbf{A} = (a_{ij})$ , so  $\mathbf{v}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$ . Expanding the wedge in each of its  $k$  slots gives one term per function  $\varphi: [k] \rightarrow [n]$ :

$$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k = \sum_{\varphi} a_{\varphi(1)1} \cdots a_{\varphi(k)k} \mathbf{e}_{\varphi(1)} \wedge \cdots \wedge \mathbf{e}_{\varphi(k)}.$$

A non-injective  $\varphi$  repeats a basis vector, so its wedge is  $\mathbf{0}$  and its term drops out. An injective  $\varphi$  is determined by its image  $I = \{i_1 < \cdots < i_k\}$  together with the permutation  $\sigma \in S_k$  defined by  $\varphi(j) = i_{\sigma(j)}$ ; conversely every pair  $(I, \sigma)$  arises from exactly one injective  $\varphi$ . For such a  $\varphi$ , Lemma 14.8.1 (b) gives

$$\mathbf{e}_{\varphi(1)} \wedge \cdots \wedge \mathbf{e}_{\varphi(k)} = \mathbf{e}_{i_{\sigma(1)}} \wedge \cdots \wedge \mathbf{e}_{i_{\sigma(k)}} = \operatorname{sgn}(\sigma) \mathbf{e}_I.$$

Grouping the terms by  $I$  therefore gives

$$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k = \sum_I \left( \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) a_{i_{\sigma(1)}1} \cdots a_{i_{\sigma(k)}k} \right) \mathbf{e}_I.$$

By Definition 6.7.1 the  $(p, q)$ -entry of  $\mathbf{A}_{I,[k]}$  is  $a_{i_p q}$ , so the inner sum is exactly the Leibniz sum of Definition 6.4.1 for the  $k \times k$  matrix  $\mathbf{A}_{I,[k]}$ , that is,  $\det \mathbf{A}_{I,[k]}$ . This proves the theorem. ■

#### Example. (A wedge of two vectors in four-dimensional space)

In  $\mathbb{Q}^4$ , let  $\mathbf{v}_1 = (1, 0, 2, 1)$  and  $\mathbf{v}_2 = (0, 1, 1, 3)$ . Write  $\mathbf{v}_1 \wedge \mathbf{v}_2$  in the basis of  $\Lambda^2 \mathbb{Q}^4$ .

*Solution.* Let  $\mathbf{A} \in M_{4 \times 2}(\mathbb{Q})$  have these two vectors as its columns, so that  $\mathbf{A}_{\{1,2\},[2]} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$  and the other five  $2 \times 2$  submatrices read off the same way. Their determinants are

$$\begin{aligned} \det \mathbf{A}_{\{1,2\}} &= 1, & \det \mathbf{A}_{\{1,3\}} &= 1, & \det \mathbf{A}_{\{1,4\}} &= 3, \\ \det \mathbf{A}_{\{2,3\}} &= -2, & \det \mathbf{A}_{\{2,4\}} &= -1, & \det \mathbf{A}_{\{3,4\}} &= 5, \end{aligned}$$

abbreviating  $\mathbf{A}_{I,[2]}$  to  $\mathbf{A}_I$ . By Theorem 14.9.5,

$$\begin{aligned}\mathbf{v}_1 \wedge \mathbf{v}_2 &= \mathbf{e}_{12} + \mathbf{e}_{13} + 3\mathbf{e}_{14} \\ &\quad - 2\mathbf{e}_{23} - \mathbf{e}_{24} + 5\mathbf{e}_{34}.\end{aligned}$$

As a check, some coordinate is non-zero, so  $\mathbf{v}_1 \wedge \mathbf{v}_2 \neq \mathbf{0}$ , so  $(\mathbf{v}_1, \mathbf{v}_2)$  is linearly independent by Theorem 14.8.7 — which is visible directly, since neither vector is a multiple of the other.

That check generalizes at once, and what comes out is a theorem of Chapter 6.

### Corollary 14.9.6 — Independence via Minors

Let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^n$  and let  $\mathbf{A} \in M_{n \times k}(F)$  have these vectors as its columns. Then  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent if and only if some  $k \times k$  minor of  $\mathbf{A}$  is non-zero.

*Proof.* By Theorem 14.8.7 the list is independent exactly when  $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k \neq \mathbf{0}$ . By Theorem 14.9.5 the coordinates of that wedge in the basis  $(\mathbf{e}_I)$  of Theorem 14.8.5 are the  $k \times k$  minors of  $\mathbf{A}$ , and a vector is  $\mathbf{0}$  exactly when all of its coordinates in a basis vanish. (For  $k > n$  both sides fail:  $\Lambda^k F^n = \{\mathbf{0}\}$  and  $\mathbf{A}$  has no  $k \times k$  submatrix.)

■

This is Theorem 6.7.1 in the case of full column rank, and the general case follows from it: rank  $\mathbf{A}$  is the largest number of independent columns, hence the largest  $k$  for which some  $k$  columns carry a non-zero  $k \times k$  minor. Chapter 6 proved it with two careful steps involving pivot columns and row rank. The wedge proves it by naming the coordinates.

## 14.9.6 Laplace expansion

Cofactor expansion also comes out of the coordinate theorem, once the wedge of  $n - 1$  vectors is available. The route is: pull the chosen column out of the wedge, recognize what is left as an element of  $\Lambda^{n-1} F^n$  whose coordinates are the  $(n - 1) \times (n - 1)$  minors, and wedge  $\mathbf{e}_i$  back on.

### Corollary 14.9.7 — Cofactor Expansion, from a Wedge

Let  $n \geq 2$ , let  $\mathbf{A} \in M_n(F)$  with minors  $M_{ij}$  as in Definition 6.6.1, and fix  $j \in [n]$ . Then

$$\det \mathbf{A} = \sum_{i=1}^n (-1)^{i+j} a_{ij} M_{ij}.$$

*Idea.* Write  $\det \mathbf{A}$  as the wedge of the columns. Linearity in the  $j$ -th slot splits it into  $n$  pieces, one per  $i$ , with  $\mathbf{e}_i$  sitting in that slot. Move  $\mathbf{e}_i$  to the front at a cost of  $(-1)^{j-1}$ ; the remaining  $n - 1$  columns form a wedge whose coordinates are the minors of the matrix with column  $j$  deleted, and wedging  $\mathbf{e}_i$  on kills every coordinate except the one whose index set misses  $i$ .

*Proof.* Let  $\mathbf{a}_1, \dots, \mathbf{a}_n$  be the columns of  $\mathbf{A}$  and  $\omega_0 = \mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_n$ . By [Theorem 14.9.3](#) applied to the operator  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ , whose matrix in the standard basis is  $\mathbf{A}$ ,

$$\mathbf{a}_1 \wedge \dots \wedge \mathbf{a}_n = (\det \mathbf{A}) \omega_0.$$

Substituting  $\mathbf{a}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$  and using linearity in slot  $j$ ,

$$(\det \mathbf{A}) \omega_0 = \sum_{i=1}^n a_{ij} \mathbf{a}_1 \wedge \dots \wedge \underset{\text{slot } j}{\mathbf{e}_i} \wedge \dots \wedge \mathbf{a}_n.$$

Fix  $i$ . Moving  $\mathbf{e}_i$  from slot  $j$  to the front means  $j - 1$  exchanges of neighbors, so by [Lemma 14.8.1 \(b\)](#) the  $i$ -th wedge above equals  $(-1)^{j-1} \varepsilon_{\mathbf{e}_i}(\eta)$ , where  $\eta = \mathbf{a}_1 \wedge \dots \wedge \widehat{\mathbf{a}_j} \wedge \dots \wedge \mathbf{a}_n \in \Lambda^{n-1} F^n$  omits the  $j$ -th column, and  $\varepsilon_{\mathbf{e}_i}$  is the map of [Lemma 14.9.1](#).

Let  $\mathbf{B} \in M_{n \times (n-1)}(F)$  be  $\mathbf{A}$  with column  $j$  deleted. By [Theorem 14.9.5](#),

$$\eta = \sum_{l=1}^n (\det \mathbf{B}_{[n] \setminus \{l\}, [n-1]}) \mathbf{e}_{[n] \setminus \{l\}},$$

since the  $(n-1)$ -element subsets of  $[n]$  are exactly the complements of single elements. Deleting row  $l$  from  $\mathbf{B}$  deletes row  $l$  and column  $j$  from  $\mathbf{A}$ , so  $\det \mathbf{B}_{[n] \setminus \{l\}, [n-1]} = M_{lj}$  by [Definition 6.6.1](#). Now apply  $\varepsilon_{\mathbf{e}_i}$ . For  $l \neq i$  the index set  $[n] \setminus \{l\}$  contains  $i$ , so  $\mathbf{e}_i \wedge \mathbf{e}_{[n] \setminus \{l\}}$  repeats  $\mathbf{e}_i$  and vanishes; for  $l = i$ , moving  $\mathbf{e}_i$  into its place costs  $i - 1$  exchanges, so  $\mathbf{e}_i \wedge \mathbf{e}_{[n] \setminus \{i\}} = (-1)^{i-1} \omega_0$  by [Lemma 14.8.1 \(b\)](#). Hence  $\varepsilon_{\mathbf{e}_i}(\eta) = (-1)^{i-1} M_{ij} \omega_0$ , and

$$(\det \mathbf{A}) \omega_0 = \sum_{i=1}^n a_{ij} (-1)^{j-1} (-1)^{i-1} M_{ij} \omega_0.$$

Since  $(-1)^{i+j-2} = (-1)^{i+j}$  and  $\omega_0 \neq \mathbf{0}$  ([Theorem 14.8.7](#)), comparing coefficients gives the stated formula. ■

This is [Theorem 6.6.1 \(a\)](#). The row version follows from it by [Theorem 6.4.5](#), exactly as in [Chapter 6](#). The gain is again structural rather than practical: the sign  $(-1)^{i+j}$ , which in [Chapter 6](#) arrived from a careful count of row and column moves, is here the price of two reversals,  $j - 1$  in one direction and  $i - 1$  in the other.

## 14.9.7 Exercises

### 14.9.7.1 A. Check your understanding

#### Exercise A1.

- (a) State the defining property of  $\Lambda^k T$ , and say why a rule given on wedges needs an argument before it defines a map.
- (b) State [Theorem 14.9.3](#).
- (c) Name one thing this section's construction of the determinant adds to [Chapter 6](#)'s, and one thing it does not.

- (d) True or false: for every  $T \in \mathcal{L}(V)$  and every  $k$ , the map  $\Lambda^k T$  is multiplication by a scalar. Justify your answer.
- (e) What are the coordinates of  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$  in  $\Lambda^k F^n$ , in the basis  $(\mathbf{e}_I)$ ?

*Solution.*

- (a)  $\Lambda^k T$  is the unique linear map  $\Lambda^k V \rightarrow \Lambda^k W$  with  $\Lambda^k T(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k) = T\mathbf{v}_1 \wedge \cdots \wedge T\mathbf{v}_k$  (Definition 14.9.1). An element of  $\Lambda^k V$  has many expressions as a sum of wedges, so a rule stated on wedges might assign two different values to one element; the universal property of Definition 14.8.2 removes the danger by producing the map from an alternating multilinear map upstairs.
- (b) If  $\dim V = n \geq 1$  and  $T \in \mathcal{L}(V)$ , then  $\dim \Lambda^n V = 1$  and  $\Lambda^n T(\omega) = (\det T)\omega$  for every  $\omega \in \Lambda^n V$ .
- (c) It adds that the determinant is the induced map on a line, so that no basis and no similarity-invariance argument is needed to define  $\det T$ , and multiplicativity becomes Theorem 14.9.2. It does not add a new value, a new method of computation, or any improvement to  $\det \mathbf{A}^\top = \det \mathbf{A}$ .
- (d) False. It is true when  $k = n = \dim V$ , and when  $k > n$  for the trivial reason that  $\Lambda^k V = \{\mathbf{0}\}$ . For  $1 \leq k < n$  it usually fails: see Example (The second exterior power of a  $3 \times 3$  matrix), where  $\Lambda^2 T$  has matrix with rows  $(8, 2, -11)$ ,  $(-1, -13, -5)$ ,  $(-20, -5, 2)$ , not a scalar matrix.
- (e) The  $k \times k$  minors of the matrix whose columns are  $\mathbf{v}_1, \dots, \mathbf{v}_k$ : the coordinate at  $\mathbf{e}_I$  is  $\det \mathbf{A}_{I,[k]}$  (Theorem 14.9.5).

### 14.9.7.2 B. Practice

#### Exercise B1: A second exterior power.

Let  $T: \mathbb{Q}^3 \rightarrow \mathbb{Q}^3$  have standard matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}.$$

Find the matrix of  $\Lambda^2 T$  in the basis  $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$ . Hence verify that  $\det(\Lambda^2 T) = (\det T)^2$ .

*Solution.* By Definition 14.9.1 and Theorem 14.9.5, the entry of  $[\Lambda^2 T]$  in row  $I$  and column  $J$  is  $\det \mathbf{A}_{I,J}$ , with rows and columns indexed by  $\{1, 2\}, \{1, 3\}, \{2, 3\}$  in that order. Computing the nine  $2 \times 2$  minors,

$$[\Lambda^2 T] = \begin{pmatrix} 1 & 3 & 6 \\ -4 & 1 & 2 \\ -2 & -6 & 1 \end{pmatrix}.$$

Its determinant is 169. Expanding  $\det \mathbf{A}$  along the first column (Theorem 6.6.1) gives  $1 \cdot (1 - 0) + 2 \cdot (6 - 0) = 13$ , and  $13^2 = 169$ .

**Exercise B2: Independence by minors.**

In  $\mathbb{Q}^4$ , let  $\mathbf{v}_1 = (1, 2, 0, 1)$ ,  $\mathbf{v}_2 = (0, 1, 1, 1)$  and  $\mathbf{v}_3 = (1, 0, -2, -1)$ . Compute the coordinates of  $\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \mathbf{v}_3$  in  $\Lambda^3 \mathbb{Q}^4$ , and hence determine whether the list is linearly independent. Justify your answer.

*Solution.* Let  $\mathbf{A} \in M_{4 \times 3}(\mathbb{Q})$  have these columns. By [Theorem 14.9.5](#) the four coordinates are the  $3 \times 3$  minors on the row sets  $\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}$ . Computing them,

$$\det \mathbf{A}_{\{1,2,3\}} = \det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{pmatrix} = 0,$$

and the same for the other three:  $\det \mathbf{A}_{\{1,2,4\}} = \det \mathbf{A}_{\{1,3,4\}} = \det \mathbf{A}_{\{2,3,4\}} = 0$ . So  $\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \mathbf{v}_3 = \mathbf{0}$ , and by [Theorem 14.8.7](#) the list is **dependent**. Indeed  $\mathbf{v}_3 = \mathbf{v}_1 - 2\mathbf{v}_2$ .

**Exercise B3: Determinant of an inverse.**

Let  $\dim V = n \geq 1$  and let  $T \in \mathcal{L}(V)$  be invertible. Using [Theorem 14.9.2](#) and [Theorem 14.9.3](#), and no formula for the determinant, prove that  $\Lambda^n T$  is invertible and that  $\det(T^{-1}) = (\det T)^{-1}$ .

*Solution.* By [Theorem 14.9.2](#),  $(\Lambda^n T)(\Lambda^n T^{-1}) = \Lambda^n(TT^{-1}) = \Lambda^n(\text{id}_V) = \text{id}$ , and symmetrically in the other order. So  $\Lambda^n T$  is invertible with inverse  $\Lambda^n T^{-1}$ . By [Theorem 14.9.3](#) these two maps are multiplication by  $\det T$  and by  $\det(T^{-1})$  on the line  $\Lambda^n V$ , and their composite is multiplication by  $\det T \cdot \det(T^{-1})$ , which equals  $\text{id}$ . Since  $\Lambda^n V \neq \{0\}$ , this forces  $\det T \cdot \det(T^{-1}) = 1$ . In particular  $\det T \neq 0$ , and  $\det(T^{-1}) = (\det T)^{-1}$ .

**14.9.7.3 C. Going deeper****Exercise C1: The trace of an exterior power.**

Let  $\mathbf{A} \in M_n(F)$  and  $1 \leq k \leq n$ , and let  $T: F^n \rightarrow F^n$  be  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ .

- Prove that the matrix of  $\Lambda^k T$  in the basis  $(\mathbf{e}_I)$  has  $(I, J)$ -entry  $\det \mathbf{A}_{I,J}$ .
- Deduce that  $\text{tr}(\Lambda^k T)$  is the sum of the  $\binom{n}{k}$  **principal**  $k \times k$  minors of  $\mathbf{A}$ , those with  $I = J$ .
- Compute  $\text{tr}(\Lambda^2 T)$  for the matrix  $\mathbf{A}$  of [Example \(The second exterior power of a  \$3 \times 3\$  matrix\)](#), and check it against the matrix found there.

*Solution.*

- Column  $J = \{j_1 < \dots < j_k\}$  of the matrix is the coordinate vector of  $\Lambda^k T(\mathbf{e}_J) = \mathbf{a}_{j_1} \wedge \dots \wedge \mathbf{a}_{j_k}$ , where  $\mathbf{a}_j = T\mathbf{e}_j$  is the  $j$ -th column of  $\mathbf{A}$  ([Definition 14.9.1](#)). By [Theorem 14.9.5](#) applied to the  $n \times k$  matrix with columns  $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}$ , the coordinate at  $\mathbf{e}_I$  is the determinant of its submatrix on rows  $I$ , which is  $\det \mathbf{A}_{I,J}$  ([Definition 6.7.1](#)).
- The trace is the sum of the diagonal entries, that is, of the entries with  $I = J$ , and by (a) those are the principal minors  $\det \mathbf{A}_{I,I}$ .

- (c) The three principal  $2 \times 2$  minors of  $\mathbf{A}$  are  $\det\begin{pmatrix} 2 & 1 \\ 0 & 4 \end{pmatrix} = 8$ ,  $\det\begin{pmatrix} 2 & 3 \\ 5 & 1 \end{pmatrix} = -13$  and  $\det\begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} = 2$ . Their sum is  $-3$ , which is the sum of the diagonal entries  $8, -13, 2$  of the matrix computed in [Example \(The second exterior power of a  \$3 \times 3\$  matrix\)](#).

### Exercise C2: Cauchy–Binet from a wedge.

Let  $m \leq n$ , let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ , and let  $T_{\mathbf{A}}: F^n \rightarrow F^m$  and  $T_{\mathbf{B}}: F^m \rightarrow F^n$  be the associated maps.

- Show that  $\Lambda^m T_{\mathbf{B}}(\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m) = \sum_J (\det \mathbf{B}_{J,[m]}) \mathbf{e}_J$ , the sum over  $m$ -element subsets  $J \subseteq [n]$ .
- Show that  $\Lambda^m T_{\mathbf{A}}(\mathbf{e}_J) = (\det \mathbf{A}_{[m],J}) \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$  for each such  $J$ .
- Hence deduce the Cauchy–Binet formula  $\det(\mathbf{AB}) = \sum_J \det \mathbf{A}_{[m],J} \det \mathbf{B}_{J,[m]}$  (Theorem 6.7.2).

*Solution.*

- $\Lambda^m T_{\mathbf{B}}(\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m) = \mathbf{B}\mathbf{e}_1 \wedge \cdots \wedge \mathbf{B}\mathbf{e}_m$  by [Definition 14.9.1](#), and the columns of  $\mathbf{B}$  are  $\mathbf{B}\mathbf{e}_1, \dots, \mathbf{B}\mathbf{e}_m$ . Apply [Theorem 14.9.5](#) in  $F^n$  with  $k = m$ .
- Writing  $J = \{j_1 < \cdots < j_m\}$ , we get  $\Lambda^m T_{\mathbf{A}}(\mathbf{e}_J) = \mathbf{A}\mathbf{e}_{j_1} \wedge \cdots \wedge \mathbf{A}\mathbf{e}_{j_m}$ , a wedge of  $m$  vectors of  $F^m$ . By [Theorem 14.9.5](#) with  $k = m = n$  there is only one index set,  $[m]$  itself, and the coefficient is the determinant of the matrix with those columns, which is  $\mathbf{A}_{[m],J}$ .
- By [Theorem 14.9.2](#),  $\Lambda^m(T_{\mathbf{A}}T_{\mathbf{B}}) = (\Lambda^m T_{\mathbf{A}})(\Lambda^m T_{\mathbf{B}})$ . The map  $T_{\mathbf{A}}T_{\mathbf{B}}$  is the operator on  $F^m$  with matrix  $\mathbf{AB}$ , so by [Theorem 14.9.3](#) the left side sends  $\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$  to  $\det(\mathbf{AB}) \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$ . By (a) and then (b), the right side sends it to  $\sum_J \det \mathbf{B}_{J,[m]} \det \mathbf{A}_{[m],J} \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$ . Comparing coefficients on the basis vector  $\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m \neq \mathbf{0}$  gives the formula.

### Exercise C3: Exterior powers of an injection.

Let  $T: V \rightarrow W$  be an **injective** linear map between finite-dimensional spaces and let  $1 \leq k \leq \dim V$ . Prove that  $\Lambda^k T$  is injective. Is the analogous statement true with “injective” replaced by “surjective”?

*Hint: choose a basis of  $V$  and use Theorem 14.8.5 in both spaces.*

*Solution.* Let  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  be a basis of  $V$ . Since  $T$  is injective,  $(T\mathbf{v}_1, \dots, T\mathbf{v}_m)$  is linearly independent in  $W$ , so it extends to a basis  $(T\mathbf{v}_1, \dots, T\mathbf{v}_m, \mathbf{w}_{m+1}, \dots)$  of  $W$  (Theorem 1.6.5). By Theorem 14.8.5, the wedges  $\mathbf{v}_{i_1} \wedge \cdots \wedge \mathbf{v}_{i_k}$  over increasing tuples form a basis of  $\Lambda^k V$ , and  $\Lambda^k T$  sends each of them to  $T\mathbf{v}_{i_1} \wedge \cdots \wedge T\mathbf{v}_{i_k}$ , which is one of the basis vectors of  $\Lambda^k W$  provided by the extended basis. So  $\Lambda^k T$  carries a basis to a linearly independent list, hence is injective.

Yes for surjective as well, and more cheaply: if  $T$  is surjective then every  $\mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_k$  equals  $T\mathbf{v}_1 \wedge \cdots \wedge T\mathbf{v}_k$  for preimages  $\mathbf{v}_i$ , so the image of  $\Lambda^k T$  contains a spanning set of  $\Lambda^k W$  (Theorem 14.8.5) and is therefore all of it.